

Quality Control Defect Analysis and Machine Learning–Based Risk Prediction in Pharmaceutical Manufacturing

Project Report Submitted by	Pranjal Gupta
Domain	Pharmaceutical Quality Control & Data Analytics
Tools & Technologies Used	Excel Python Machine Learning Power BI
Techniques Applied	Data Cleaning • Exploratory Data Analysis (EDA) • Logistic Regression • KPI Design • Dashboarding • Conditional Formatting
Project Difficulty Level	Intermediate

Dataset Availability

The cleaned dataset used for this project is publicly available in the GitHub repository:

Dataset Link: https://raw.githubusercontent.com/pranjalgupta4482-ui/Pharma-QC-Defect-Analysis/refs/heads/main/data/defects_data_cleaned_final_csv.csv

About Dataset

The dataset used in this project represents **quality control defect records from a pharmaceutical manufacturing environment**. The data is designed to simulate real-world quality inspection scenarios, where defects are identified, classified, and assessed based on severity and repair cost.

The dataset was prepared and cleaned before analysis to ensure consistency, accuracy, and suitability for both exploratory data analysis and machine learning modelling.

Defect Dataset – Column Description

The main defect dataset includes the following attributes:

- **defect_id**: Unique identifier assigned to each defect record
- **product_id**: Identifier for the product associated with the defect
- **defect_type**: Category of defect (e.g., structural, cosmetic, contamination)
- **defect_date**: Date on which the defect was identified during inspection
- **defect_location**: Area or stage of production where the defect occurred
- **severity**: Classification of defect severity (Low / High)
- **inspection_method**: Method used to detect the defect (e.g., visual inspection, manual testing)
- **repair_cost**: Cost incurred to repair or rework the defect

All monetary values related to repair cost were stored as **numeric values** to support accurate analysis and were later formatted as currency at the visualization stage.



Machine Learning Prediction Dataset

The machine learning prediction dataset includes:

- **Actual_Severity**: True defect severity value from the test dataset
- **Predicted_Severity**: Severity predicted by the Logistic Regression model

This dataset was generated after model training and evaluation and was later integrated into Power BI to enable **ML-based risk visualization**.

Summary

The dataset provides a structured foundation for analyzing defect trends, evaluating quality risks, and applying machine learning techniques to predict defect severity. Separating raw defect data and ML predictions ensures data integrity and follows standard analytics best practices.

Dataset Files Used

This project utilizes **two separate datasets**:

1. **Defect Dataset (defects_data_cleaned.csv)**

This dataset contains detailed information about individual quality defects observed during inspections.

2. **Machine Learning Prediction Dataset (ml_predictions.csv)**

This dataset contains the actual and predicted defect severity values generated by the machine learning model and is used for dashboard integration.

Problem Statement

Pharmaceutical manufacturing operates under strict quality standards to ensure product safety, regulatory compliance, and patient well-being. Despite established quality control procedures, manufacturing defects can still occur at various stages of production. These defects may vary in type, location, and severity, and can result in increased repair costs, production delays, and potential quality risks.

Traditional quality control processes often rely on manual inspection and historical analysis, which may not be sufficient to proactively identify high-risk defects. The lack of predictive insight makes it

difficult for quality teams to prioritize critical issues and allocate resources effectively.

Therefore, there is a need for a data-driven approach that can analyse historical defect data, uncover meaningful patterns, and predict defect severity in advance. Such an approach can help organizations reduce quality risks, control repair costs, and improve overall manufacturing efficiency.

This project aims to address this challenge by applying data analytics and machine learning techniques to quality control defect data and presenting actionable insights through an interactive Power BI dashboard.

Project Overview

This project focuses on analysing quality control defect data from a pharmaceutical manufacturing environment to identify defect patterns, assess severity levels, and evaluate the impact on repair costs. Using Python-based data analytics and machine learning techniques, a Logistic Regression model was developed to predict high-risk defects. The results were further visualized through an interactive Power BI dashboard to support data-driven quality monitoring and decision-making.

Data Cleaning & Preprocessing

Data cleaning and preprocessing were performed to ensure the dataset was accurate, consistent, and suitable for analysis and machine learning. The defect date column was standardized into a proper date format, and non-numeric characters were removed from the repair cost field to enable numerical analysis. Missing values were handled appropriately, and categorical variables were encoded to convert them into a machine-learning-friendly format. These steps ensured reliable analysis and improved model performance.

https://raw.githubusercontent.com/pranjalgupta4482-ui/Pharma-QC-Defect-Analysis/refs/heads/main/defects_data_cleaned_csv.csv

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand defect distribution, severity patterns, and repair cost trends. Visual analysis helped identify key defect types, high-risk locations, and the relationship between defect severity and repair cost. These insights guided feature selection and supported informed decision-making for machine learning modelling and dashboard design.

Machine Learning Methodology

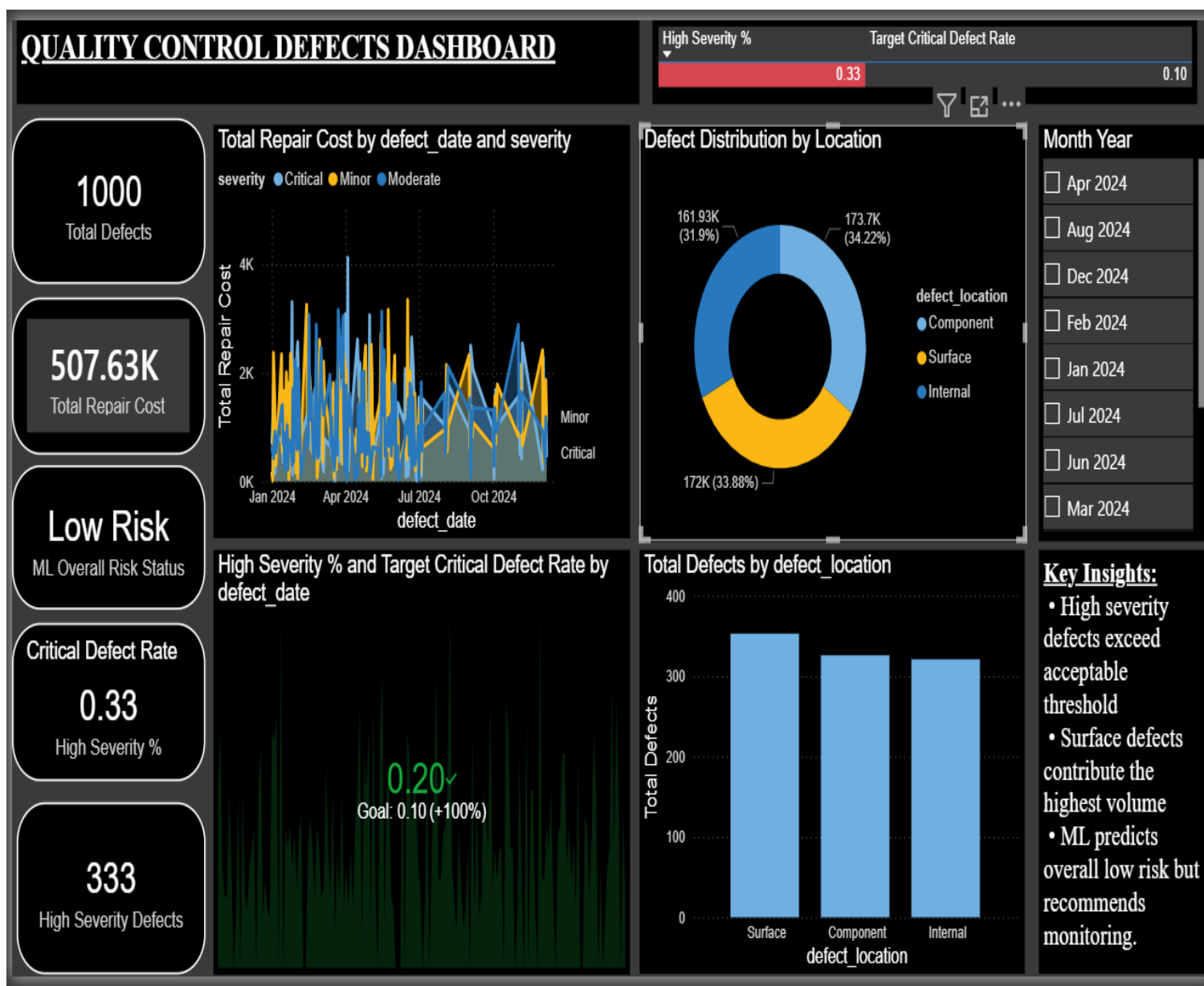
A machine learning approach was applied to predict defect severity based on historical quality control data. Logistic Regression was selected as the classification model due to its simplicity, interpretability, and suitability for binary classification problems. Categorical variables were encoded, irrelevant fields such as date were excluded, and missing values were handled to ensure the model received clean numerical input.

Model Evaluation

The trained Logistic Regression model was evaluated using standard classification metrics. A confusion matrix and classification report were used to assess model performance in terms of accuracy, precision, recall, and F1-score. The evaluation results indicate that the model is capable of distinguishing between high-severity and low-severity defects with reasonable accuracy.

Power BI Dashboard

An interactive Power BI dashboard was developed to visualize defect trends, repair cost impact, and machine learning–based risk insights. The dashboard includes key performance indicators such as total defects, total repair cost, defect distribution by severity and location, and overall risk classification derived from ML predictions. The dashboard enables stakeholders to monitor quality performance and identify high-risk areas effectively.



Key Insights

- High-severity defects contribute significantly to overall repair costs
- Certain defect locations and inspection methods show higher risk patterns
- Machine learning predictions help identify potential high-risk defects early
- Data-driven visualization improves quality monitoring and decision-making

Conclusion

This project demonstrates the effective application of data analytics and machine learning techniques in pharmaceutical quality control. By integrating Python-based analysis with Power BI visualization, the project provides actionable insights into defect patterns and severity risks. The approach supports proactive quality management and highlights the value of predictive analytics in manufacturing environments.

